



Deal with insufficient data

Every analyst has been in a situation where there is insufficient data to help with their business objective. Considering how much data is generated every day, it may be hard to believe, but it's true. So let's discuss what you can do when you have insufficient data. We'll cover how to set limits for the scope of your analysis and what data you should include. At one point, I was a data analyst at a support center. Every day, we received customer questions, which were logged in as support tickets. I was asked to forecast the number of support tickets coming in per month to figure out how many additional people we needed to hire.

It was very important that we had sufficient data spanning back at least a couple of years because I had to account for year-to-year and seasonal changes. If I just had the current year's data available, I wouldn't have known that a spike in January is common and has to do with people asking for refunds after the holidays. Because I had sufficient data, I was able to suggest we hire more people in January to prepare. Challenges are bound to come up, but the good news is that once you know your business objective, you'll be able to recognize whether you have enough data. And if you don't, you'll be able to deal with it before you start your analysis. Now, let's check out some of those limitations you might come across and how you can handle different types of insufficient

data. Say you're working in the tourism industry, and you need to find out which travel plans are searched most often.

If you only use data from one booking site, you're limiting yourself to data from just one source. Other booking sites might show different trends that you would want to consider for your analysis. If a limitation like this impacts your analysis, you can stop and go back to your stakeholders to figure out a plan. If your data set keeps updating, that means the data is still incoming and might not be complete. So if there's a brand new tourist attraction that you're analyzing interest and attendance for, there's probably not enough data for you to determine trends. For example, you might want to wait a month to gather data. Or you can check in with the stakeholders and ask about adjusting the objective. For example, you might analyze trends from week to week instead of month to month. You could also base your analysis on trends over the past three months and say, "Here's what attendance at the attraction for month four could look like." You might not have enough data to know if this number is too low or too high. But you would tell stakeholders that it's your best estimate based on the data that you currently have. On the other hand, your data could be older and no longer be relevant. Outdated data about customer satisfaction won't include the most recent responses. So you'll be relying on the ratings for hotels or vacation rentals that might no longer be accurate. In this case, your best bet might be to find a new data set to work with. Data that's geographically-limited could also be unreliable. If your company is global, you wouldn't want to use data limited to travel in just one country. You would want a data set that includes all countries. So that's just a few of the most common limitations you'll come across and some ways you can address them. **You can identify trends with the available**

data or wait for more data if time allows; you can talk with stakeholders and adjust your objective; or you can look for a new data set. The need to take these steps will depend on your role in your company and possibly the needs of the wider industry.

But learning how to deal with insufficient data is always a great way to set yourself up for success. Your data analyst powers are growing stronger. And just in time. After you learn more about limitations and solutions, you'll learn about statistical power, another fantastic tool for you to use. See you soon!